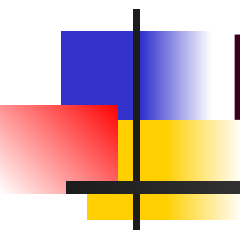
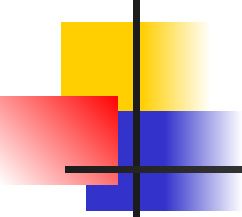


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NEUROMUSCULAR JUNCTION

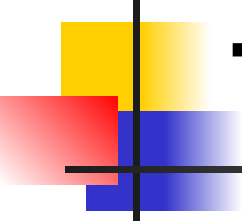
Dr Shahneela Siraj
Associate Professor
DIKIOHS ,DUHS.



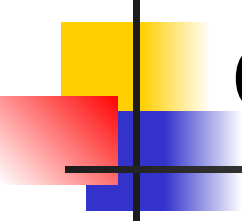
By the end of this lecture, students should be able to:

- Define Synapse and classify it
- Describe types of synaptic transmission
- Understand the steps involved in chemical transmission
- List the components of the neuromuscular junction (NMJ)
- Describe the molecular events of neurotransmission at the NMJ
- Understand end-plate potential
- Differentiate the electrical and mechanical events at the NMJ.

SYNAPTIC AND NEUROMUSCULAR TRANSMISSION



- A **synapse** is a site where information is transmitted from one cell to another. The information can be transmitted either electrically (electrical synapse) or via a chemical transmitter (chemical synapse).
- **Synaptic Transmission:** The process underlying this cell-to-cell transfer of electrical signals
- **The synaptic delay** is the time required for the multiple steps in chemical neurotransmission to occur.



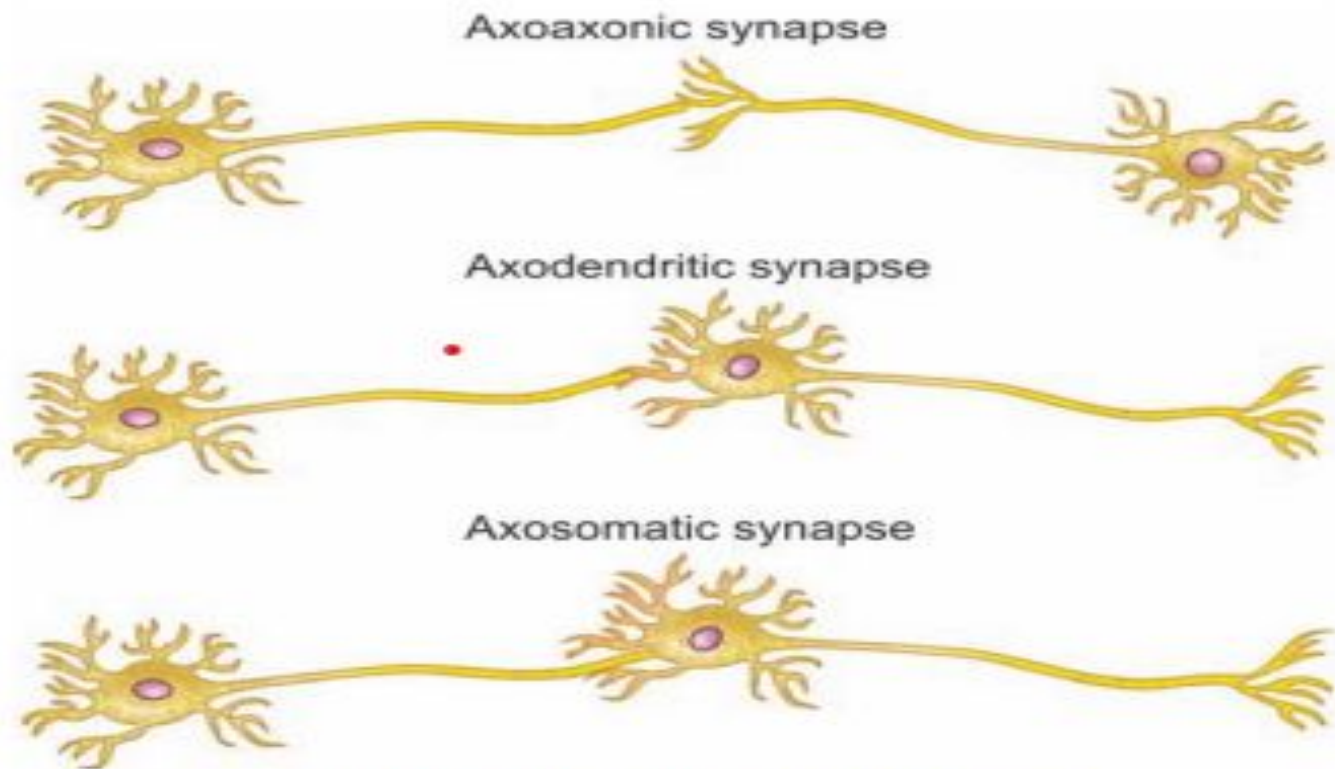
Classification Of Synapse

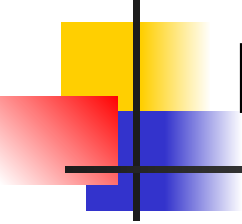
▣ **ANATOMICAL CLASSIFICATION**

Usually synapse is formed by axon of one neuron ending on the cell body, dendrite or axon of the next neuron. Depending upon ending of axon, synapse is classified into three types:

- ▣ 1. **Axoaxonic synapse** in which axon of one neuron terminates on axon of another neuron.
- ▣ 2. **Axodendritic synapse** in which the axon of one neuron terminates on dendrite of another neuron.
- ▣ 3. **Axosomatic synapse** in which axon of one neuron ends on soma (cell body) of another neuron.

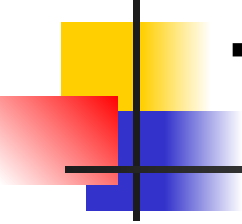
Anatomical Synapse





FUNCTIONAL CLASSIFICATION

- Functional classification of synapse is on the basis of mode of impulse transmission. According to this, synapse is classified into two categories:
- 1. Electrical synapse
- 2. Chemical synapse.

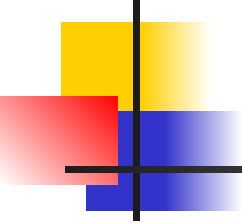


Types of Synapses

□ **Electrical Synapses**

Allow current to flow from one excitable cell to the next via low resistance pathways between the cells called gap junctions.

- Gap junctions are found in cardiac muscle and in some types of smooth muscle and account for the very fast conduction in these tissues. For example, rapid cell-to-cell conduction occurs in cardiac ventricular muscle, in the uterus, and in the bladder, allowing cells in these tissues to be activated simultaneously and ensuring that contraction occurs in a coordinated manner.

- 
-
- **Chemical Synapses** In chemical synapses, there is a gap between the presynaptic cell membrane and the postsynaptic cell membrane, known as the **synaptic cleft**.
 - Information is transmitted across the synaptic cleft via a neurotransmitter, a substance that is released from the presynaptic terminal and binds to receptors on the postsynaptic terminal.
 - In contrast to electrical synapses, neurotransmission
 - across chemical synapses is unidirectional (from presynaptic cell to postsynaptic cell).

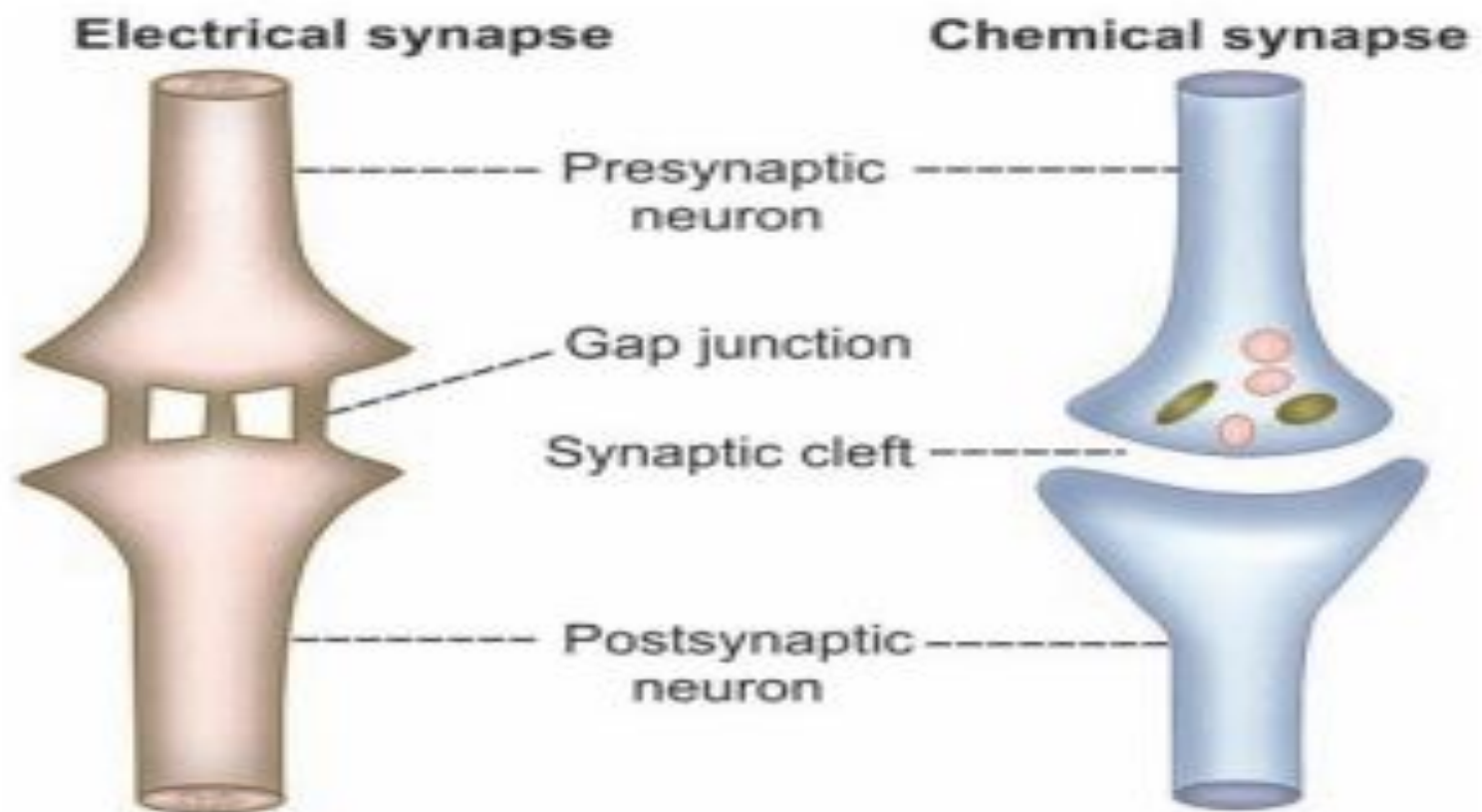
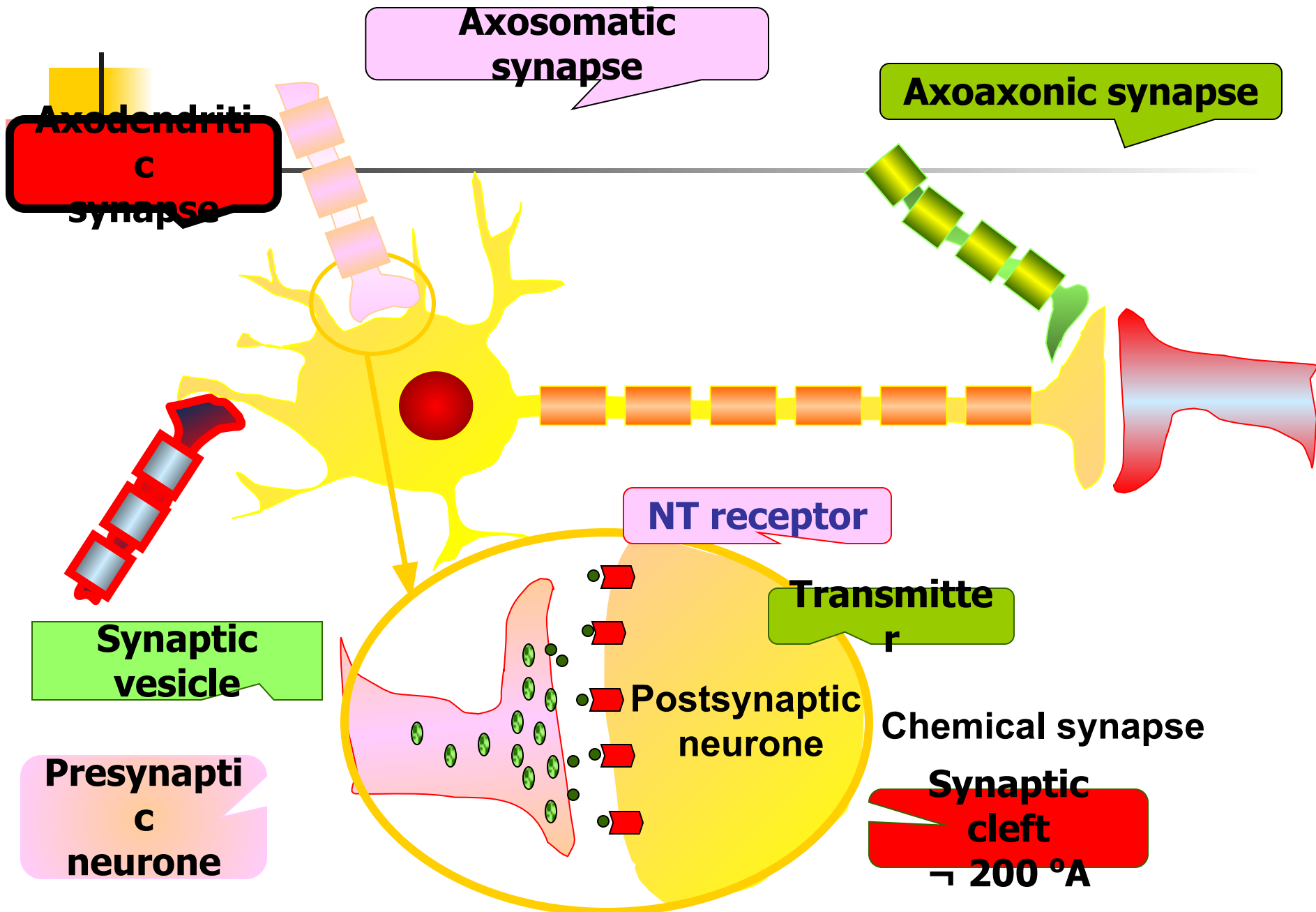


FIGURE 140.2: Electrical and chemical synapse

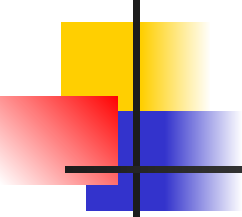
Neuron:Neuron



Synaptic Input—Excitatory and Inhibitory Postsynaptic Potentials

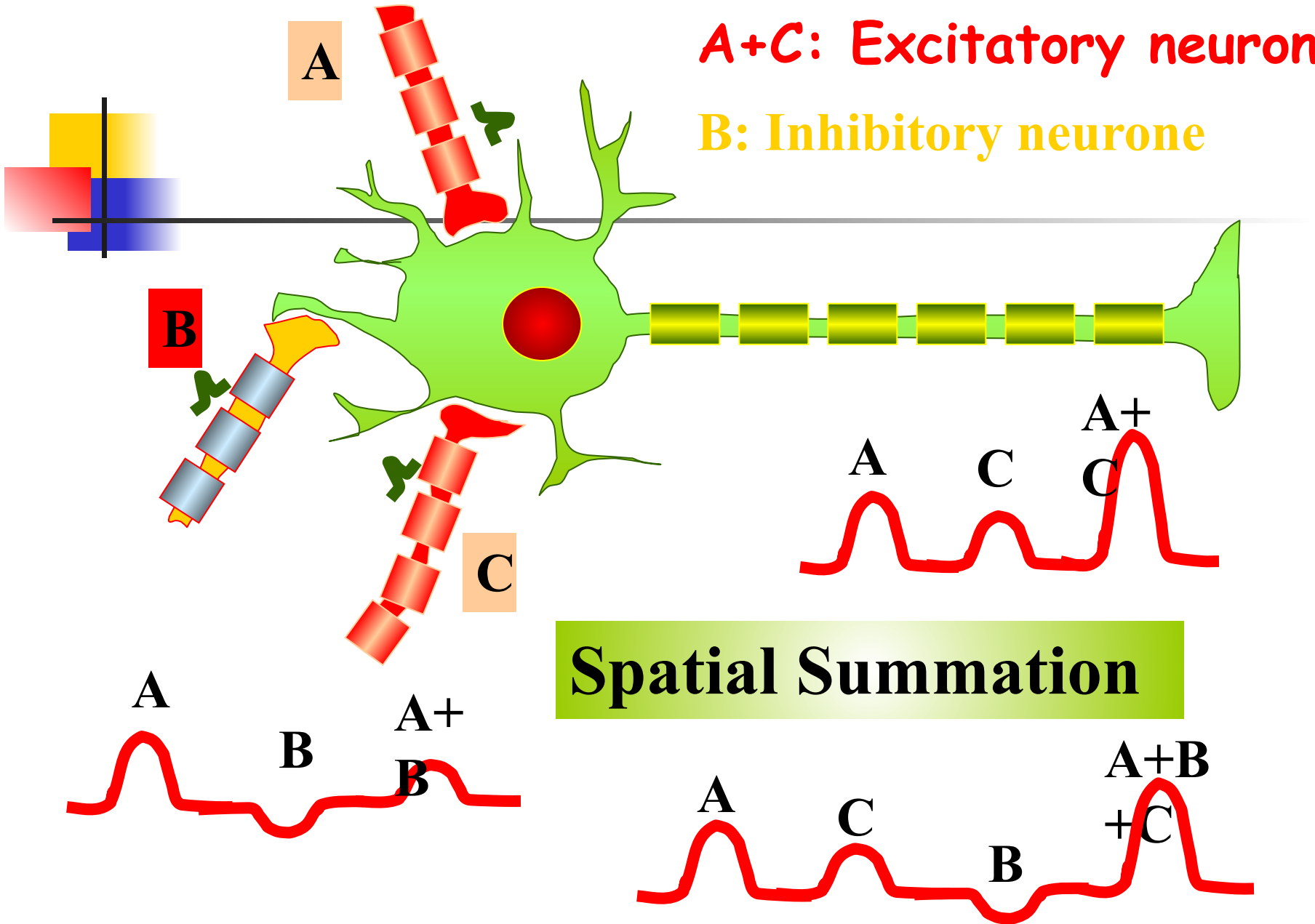
- The many-to-one synaptic arrangement is a common configuration in which many presynaptic cells converge on a single postsynaptic cell, with the inputs being either excitatory or inhibitory.
- The postsynaptic cell integrates all the converging information, and if the sum of the inputs is sufficient to bring the postsynaptic cell to threshold, it will then fire an action potential.
- **(EPSPs)** are synaptic inputs that depolarize the postsynaptic cell, bringing the membrane potential closer to threshold and closer to firing an action potential. Produced by opening Na^+ and K^+ channels.
- **(IPSPs)** are synaptic inputs that hyperpolarize the postsynaptic cell, taking the membrane potential away from threshold and farther from firing an action potential. produced by opening Cl^- channels..

Integration of Synaptic Information

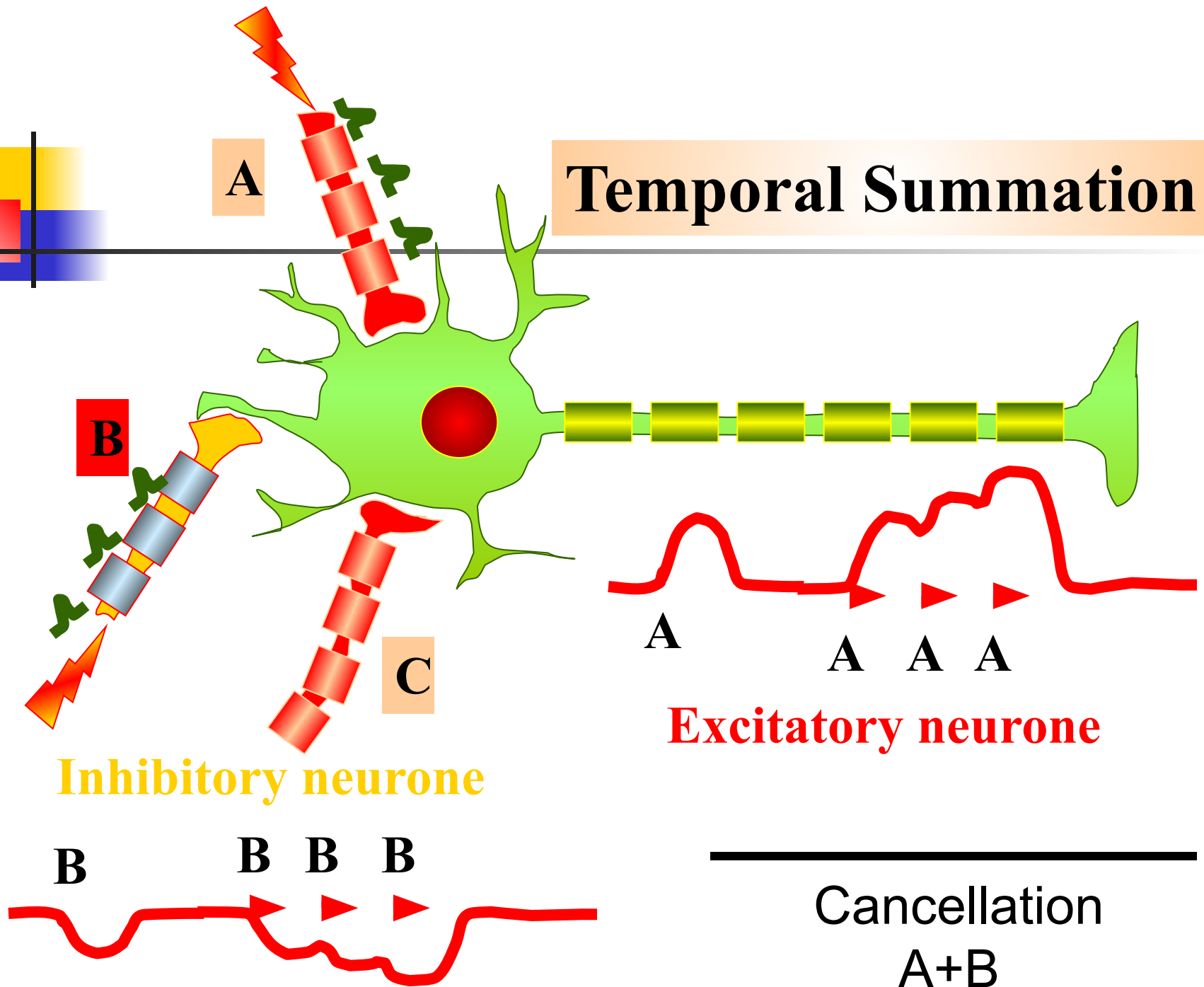
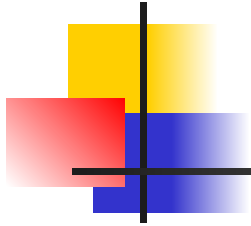
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- The presynaptic information that arrives at the synapse may be integrated in one of two ways, spatially or temporally.
 - **Spatial summation** occurs when two or more presynaptic inputs arrive at a postsynaptic cell simultaneously. If both inputs are excitatory, they will combine to produce greater depolarization than either input would produce separately. If one input is excitatory and the other is inhibitory, they will cancel each other out. Spatial summation may occur, even if the inputs are far apart on the nerve cell body, because EPSPs and IPSPs are conducted so rapidly over the cell membrane.
 - **Temporal summation** occurs when two presynaptic inputs arrive at the postsynaptic cell in rapid succession. Because the inputs overlap in time, they summate.

A+C: Excitatory neurone

B: Inhibitory neurone

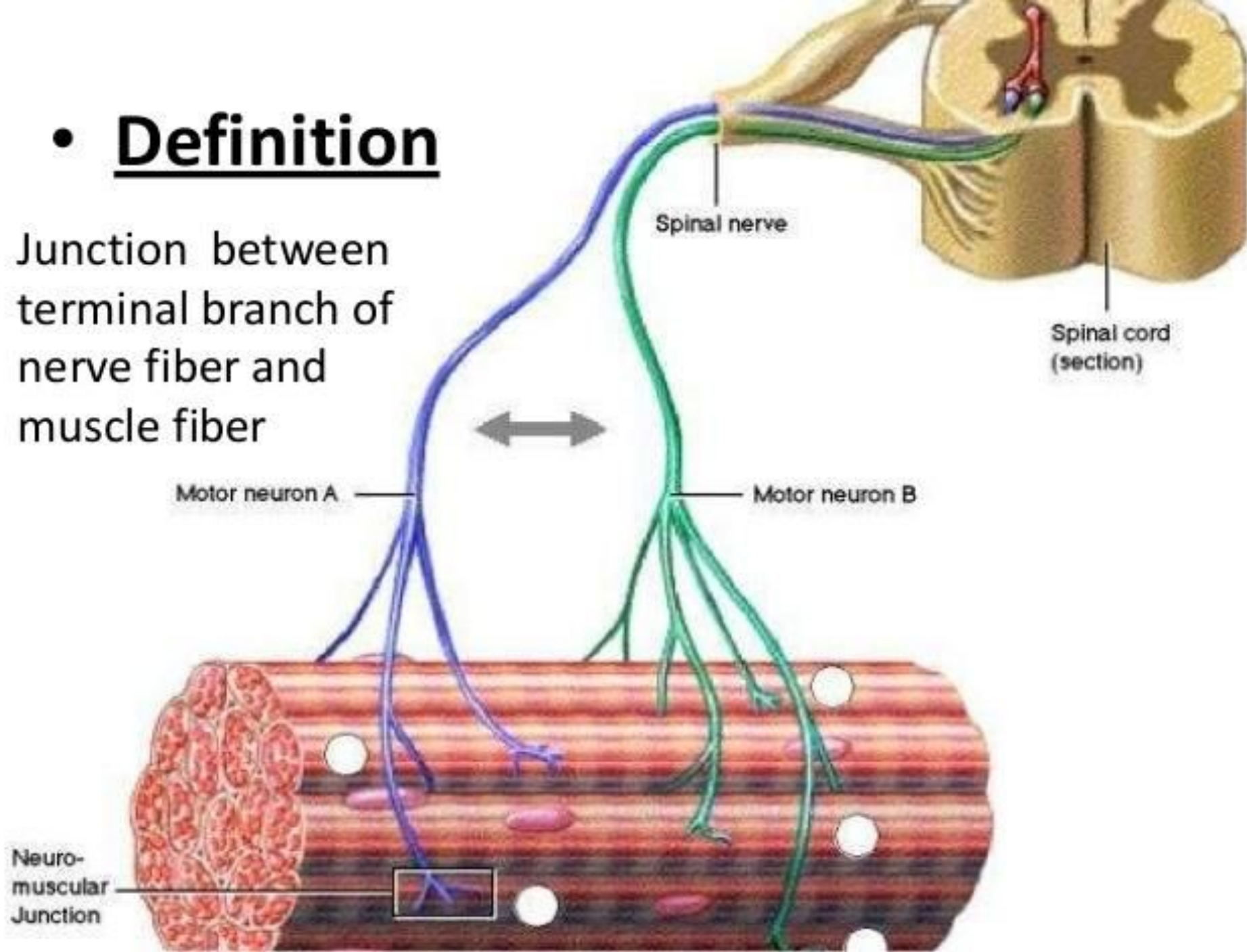


Temporal Summation

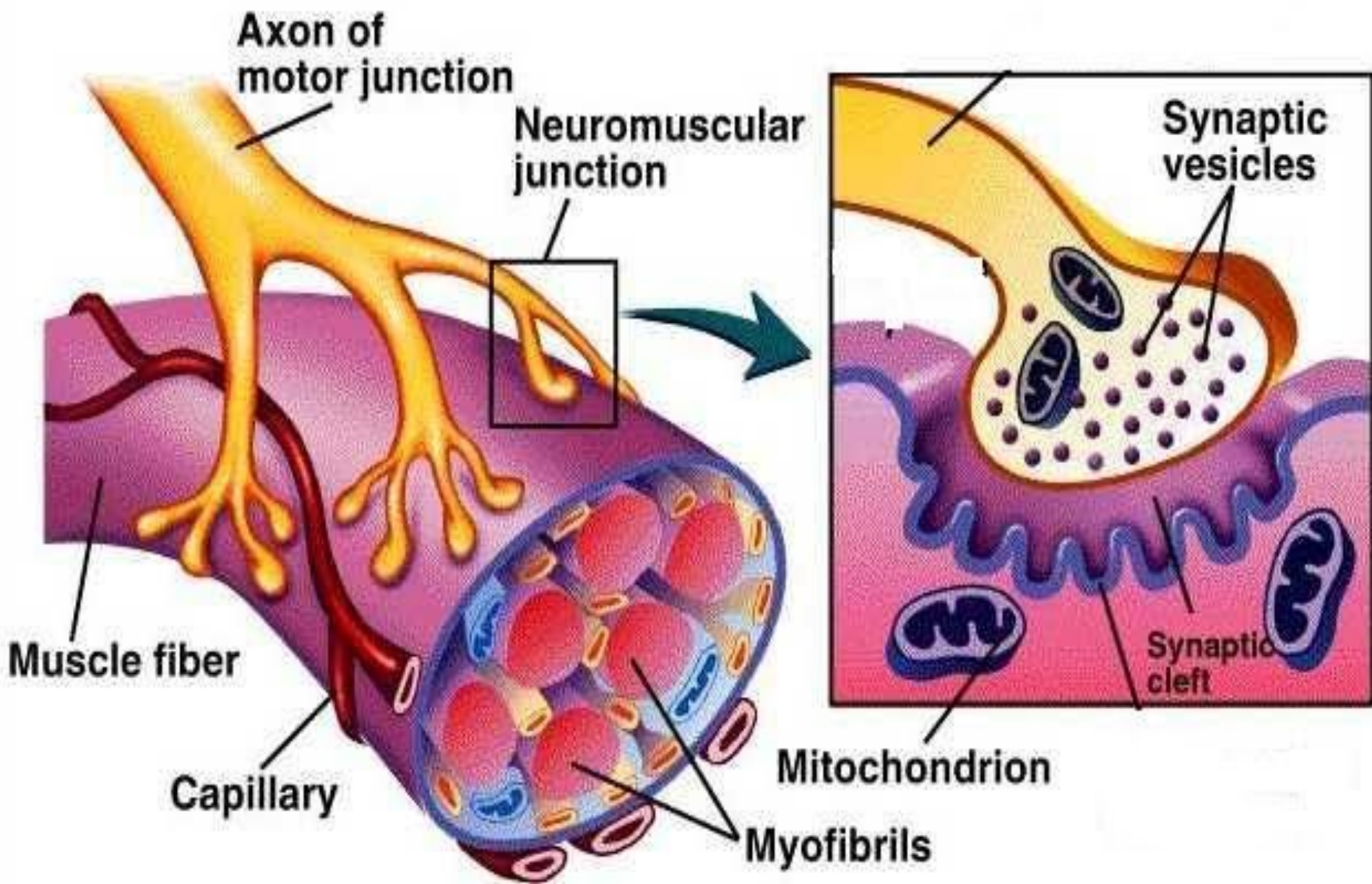


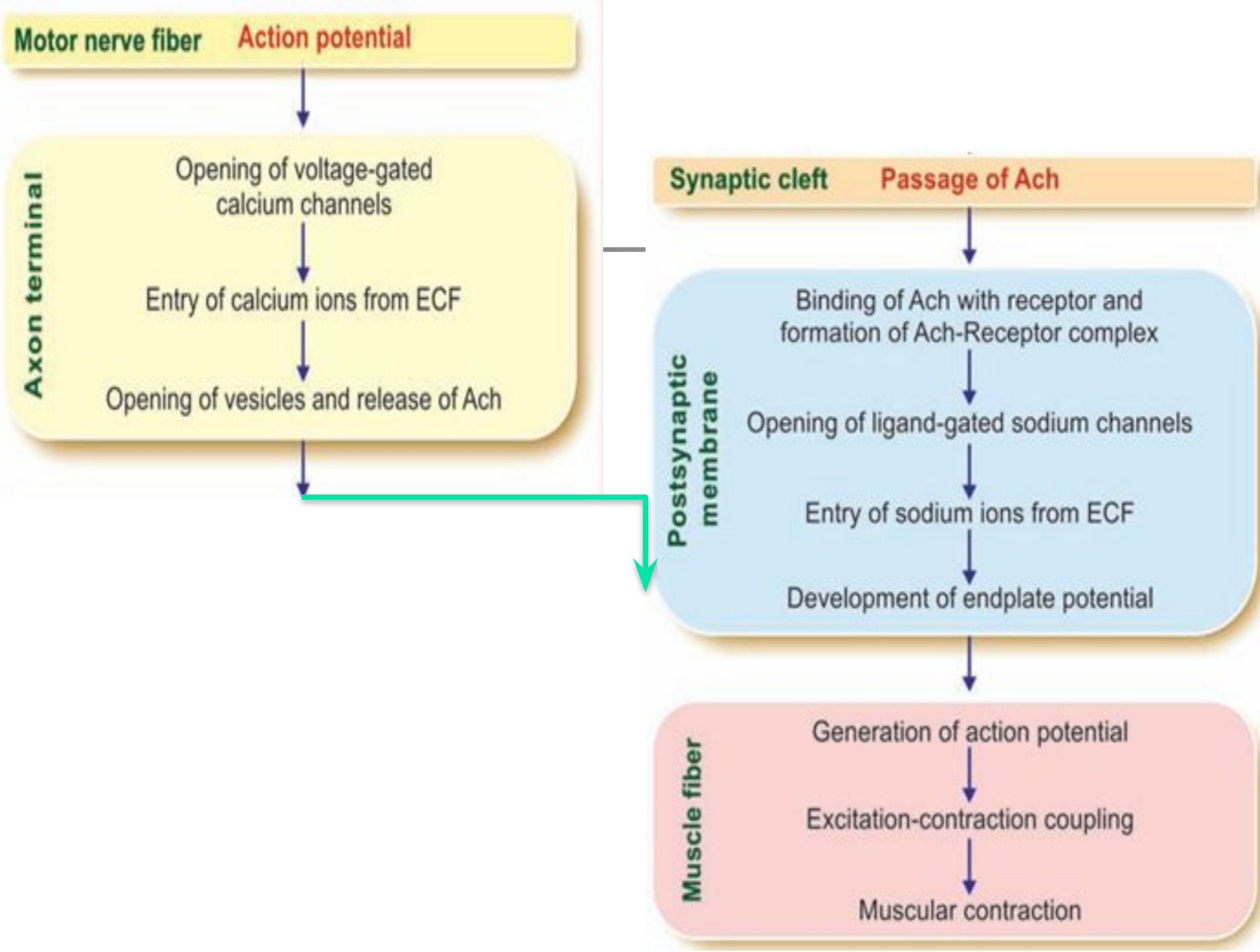
- **Definition**

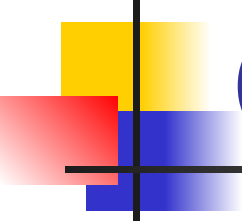
Junction between
terminal branch of
nerve fiber and
muscle fiber



Neuromuscular Junction







Neuromuscular junction

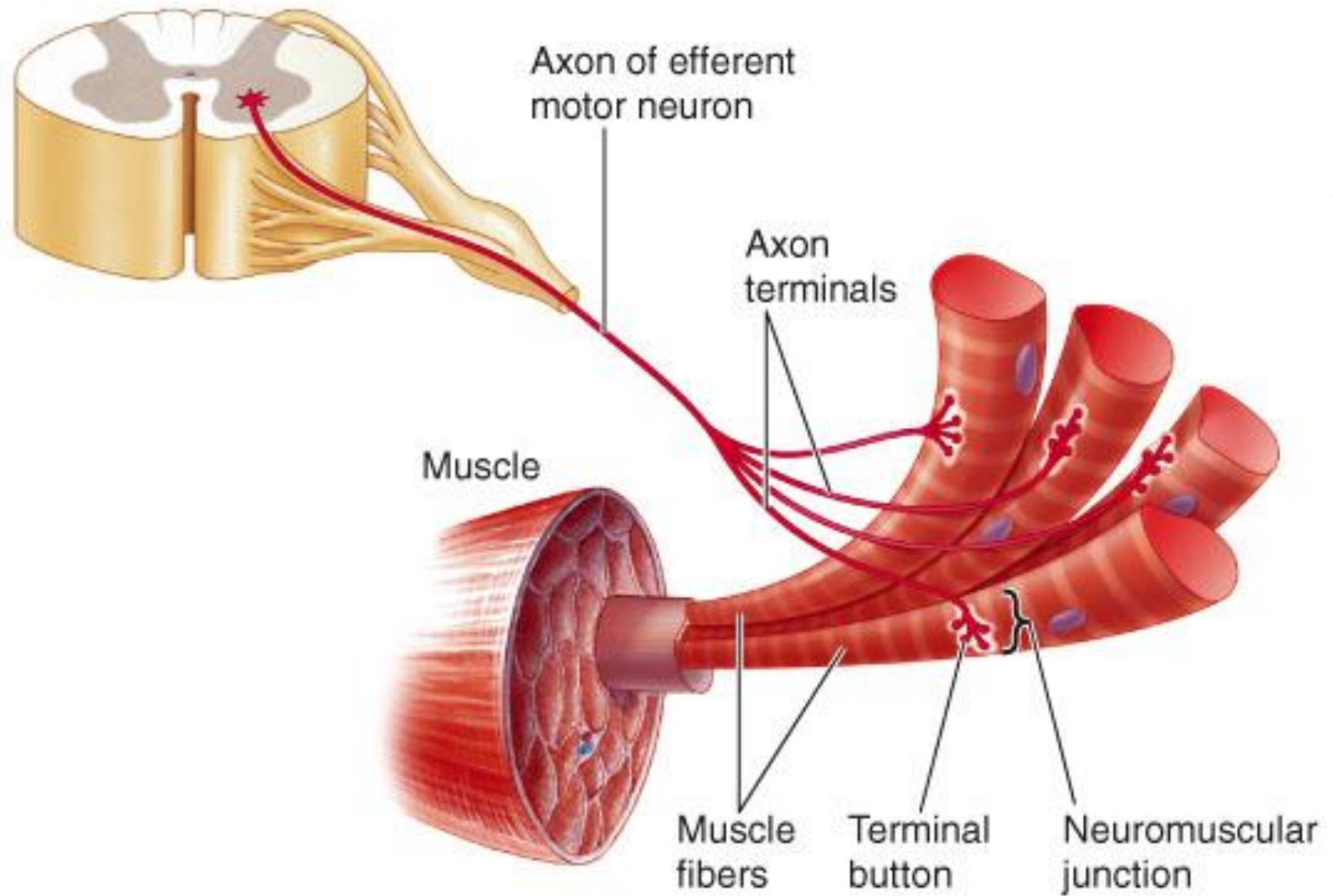
(example of chemical synapse)

- ▣ **Neuromuscular junction** : the synapse between motor neuron and muscle fiber is called the neuromuscular junction.
- ▣ **Motor neurons** : are the nerves that innervate muscle fibers .
- ▣ **Motor unit** : single motor neuron and the muscle fibers it innervate.

Physiologic anatomy of N.M junction (continued)

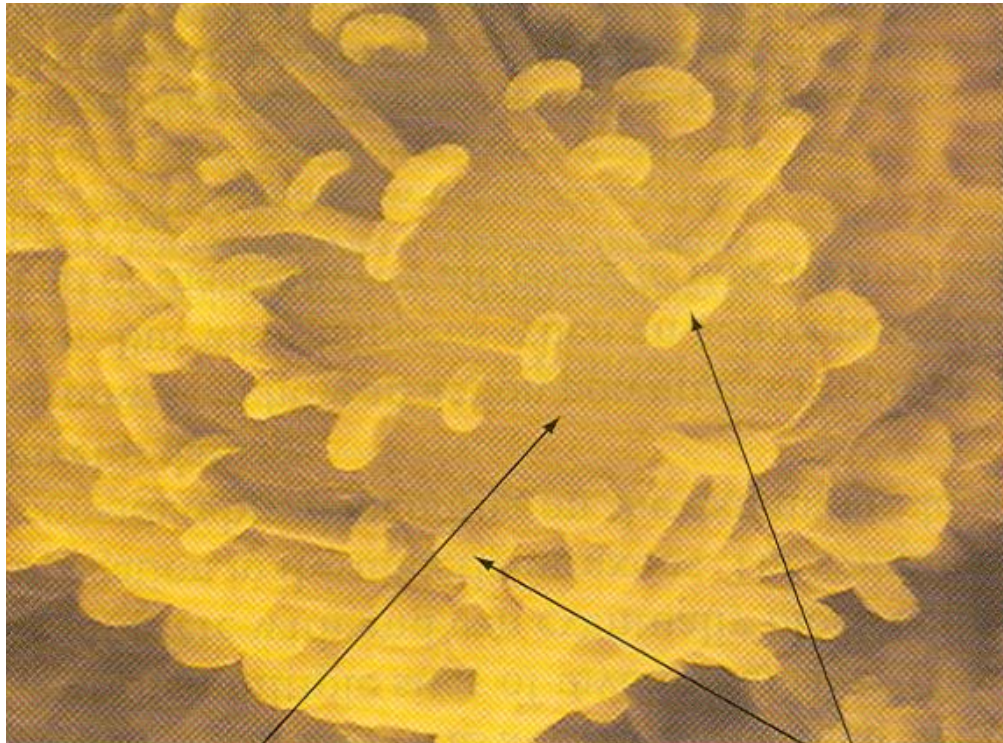
- As axon approaches muscle , it divides into many terminal branches and loses its myelin sheath.
- Each of these axon terminal forms special junction ,a neuromuscular junction with one or more muscle fiber.

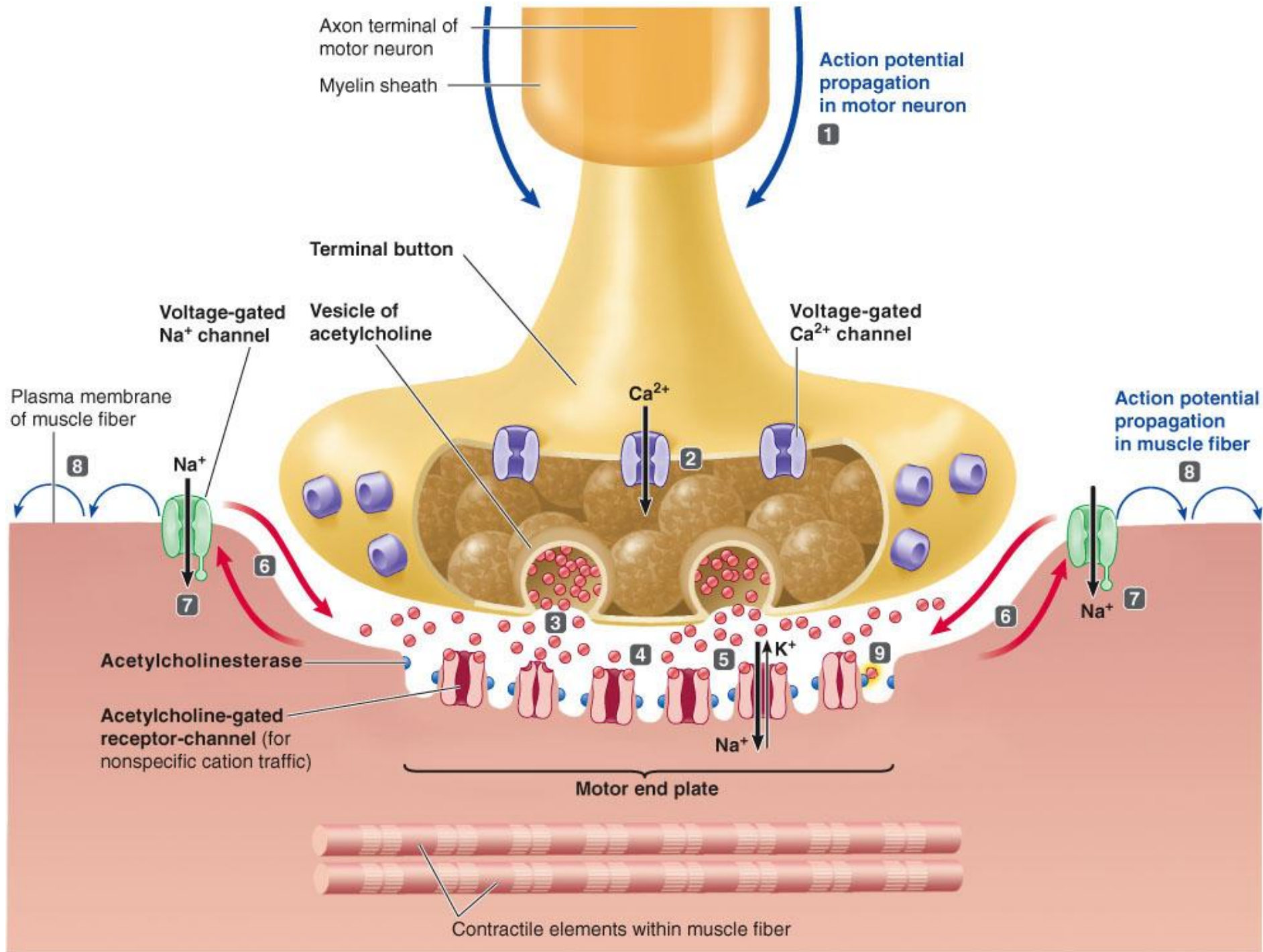
Spinal cord (section)



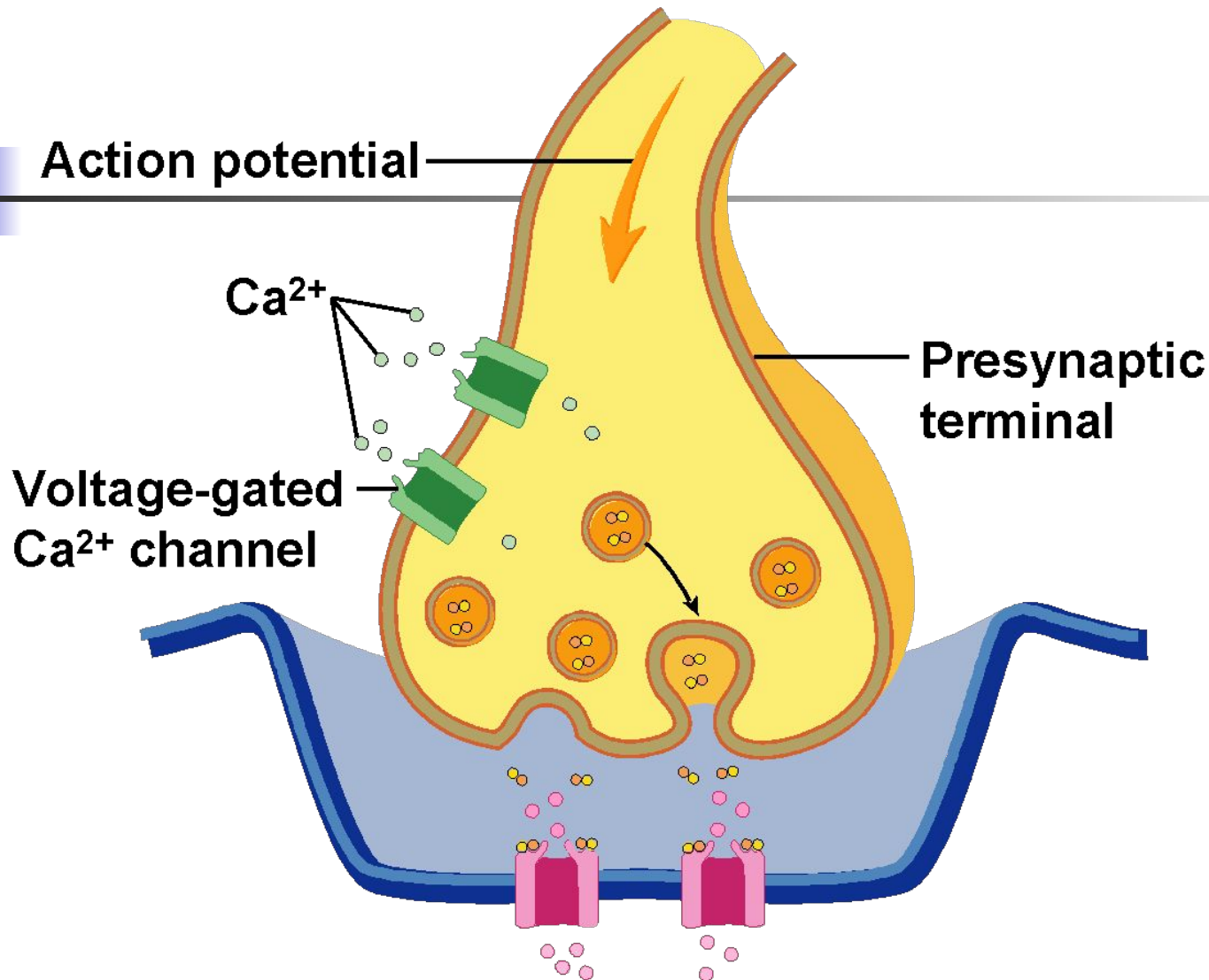
Physiologic anatomy of N.M junction (continued)

- The axon terminal is enlarged into a knoblike structure, the terminal button, which fits into shallow depression in underlying muscle fiber.





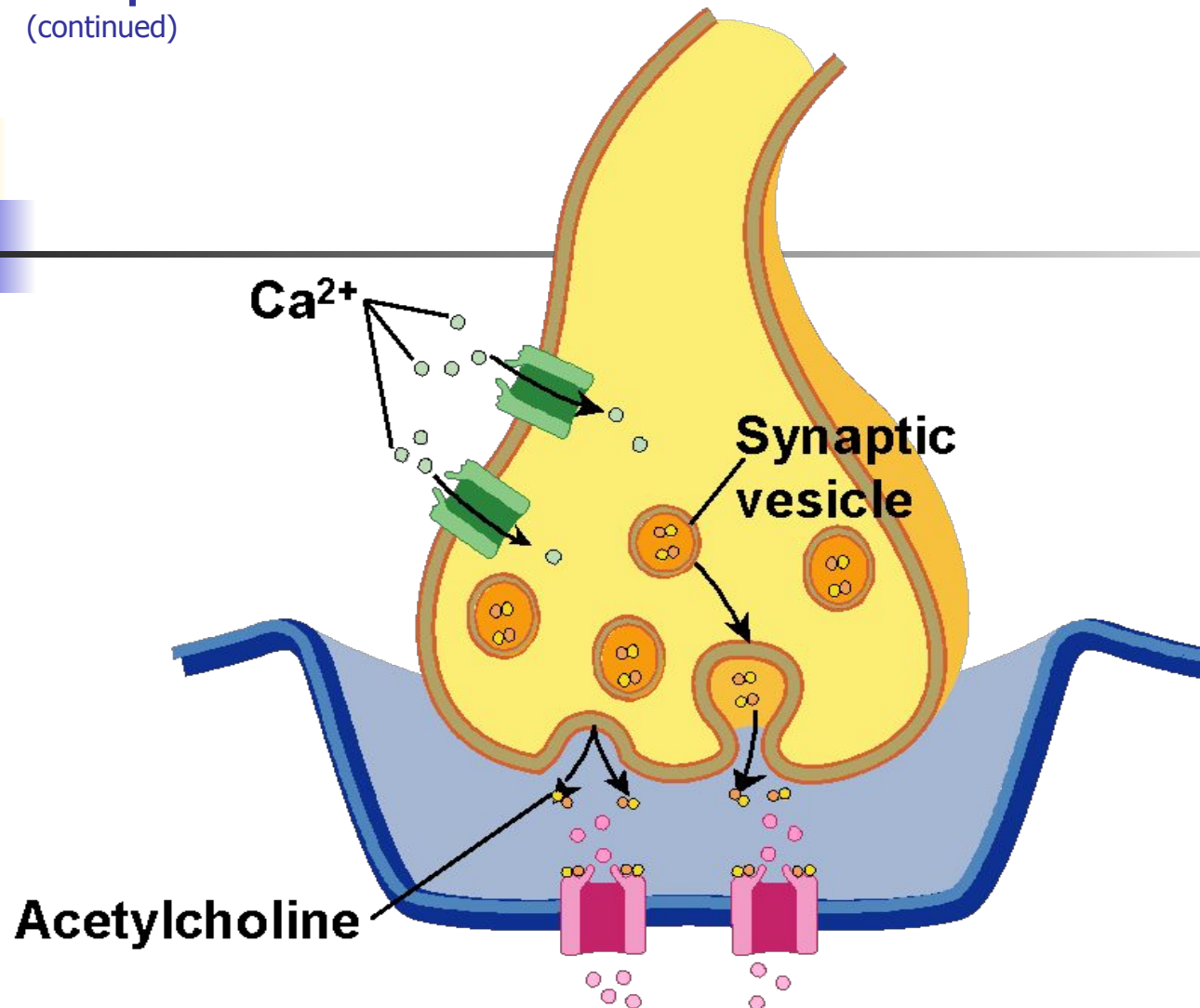
Sequence Of Events At Neuromuscular Junction



Action potentials arriving at the presynaptic terminal cause voltage-gated Ca^{2+} channels to open.

Sequence Of Events At Neuromuscular Junction

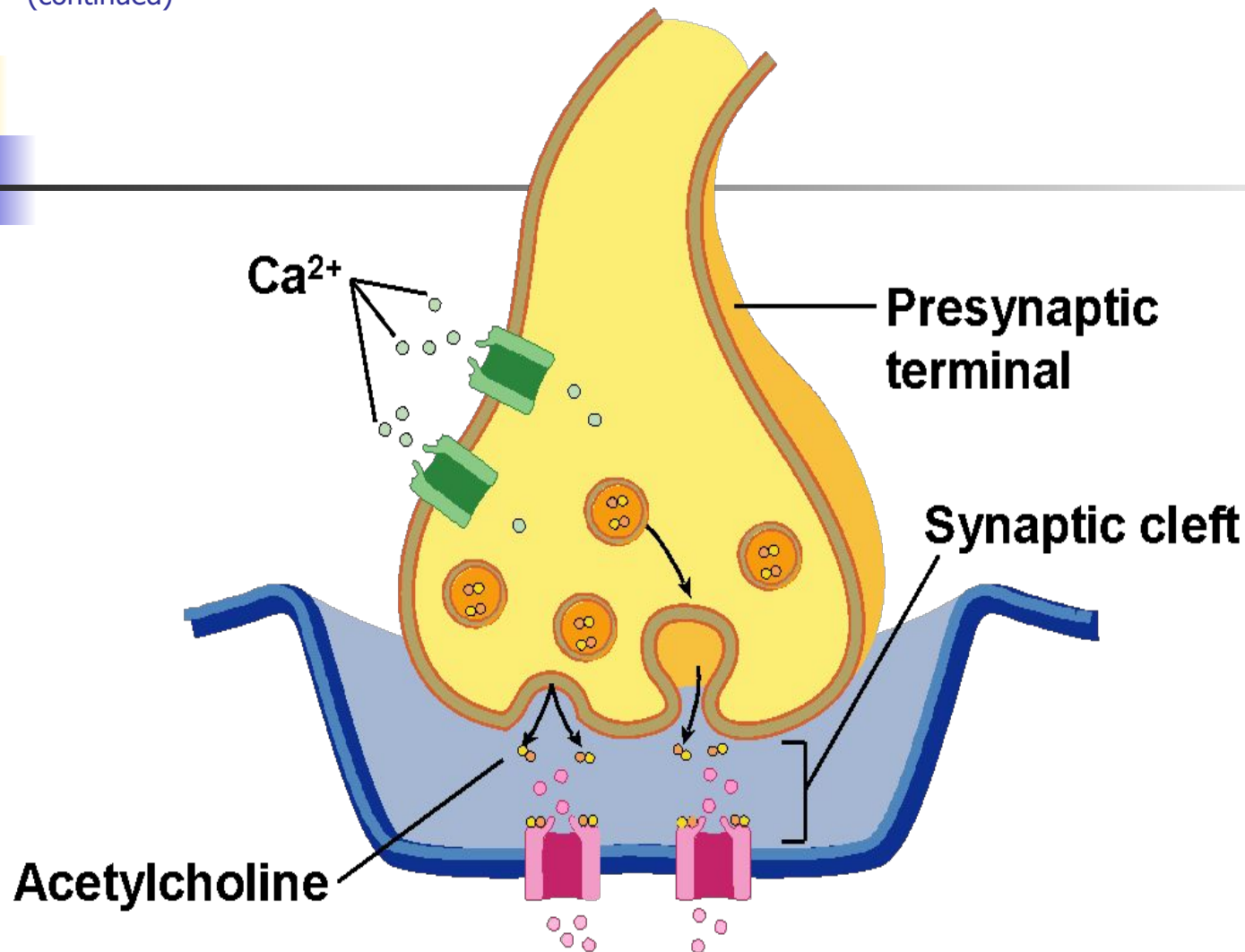
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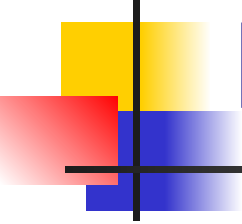
Ca^{2+} uptake into the terminal causes release of the neurotransmitter acetylcholine into synaptic cleft, which has been synthesized and stored into synaptic vesicles

Sequence Of Events At Neuromuscular Junction

(continued)



Ach travels across the synaptic cleft to postsynaptic membrane which is also known as motor end plate.

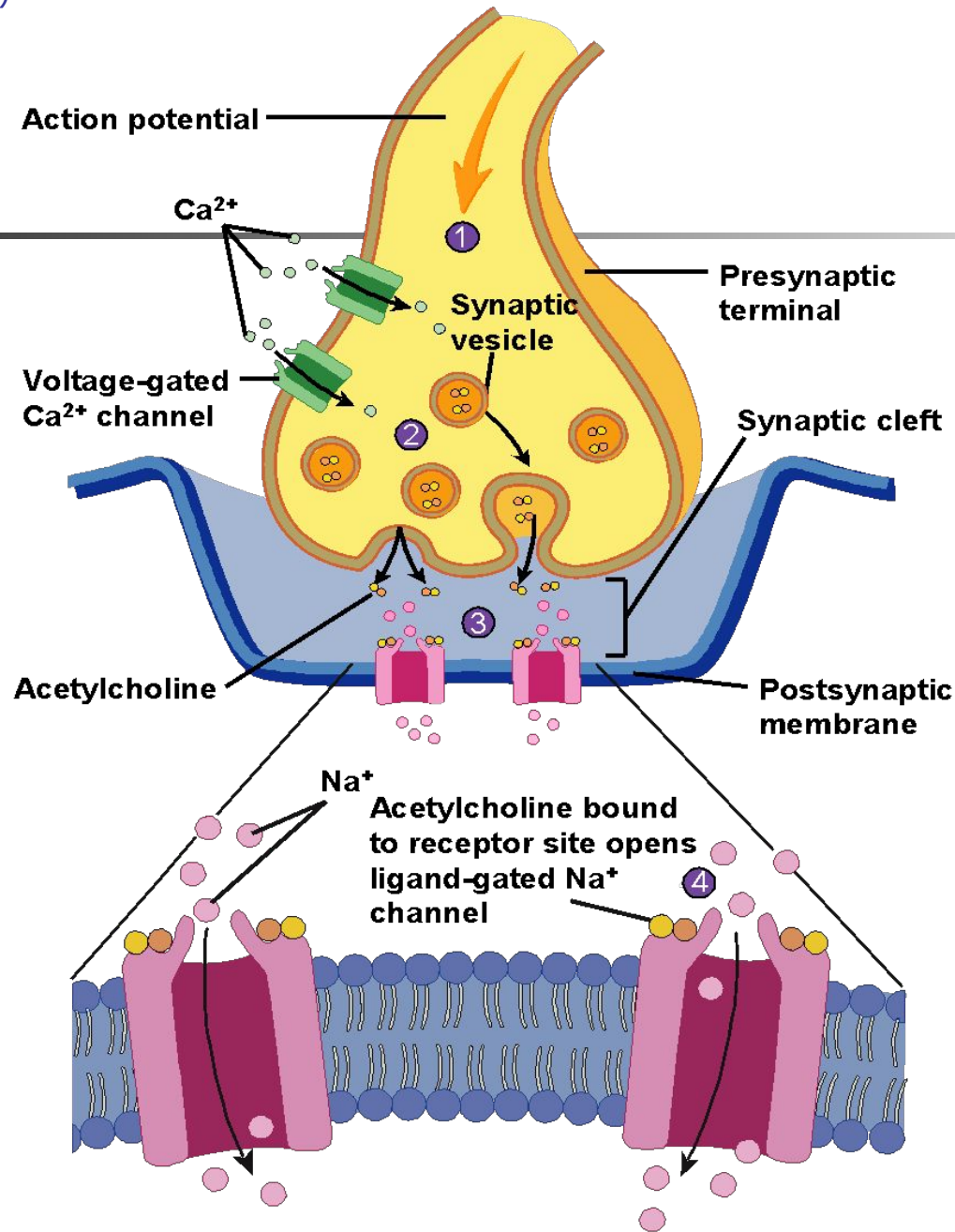


Sequence Of Events At Neuromuscular Junction (continued)

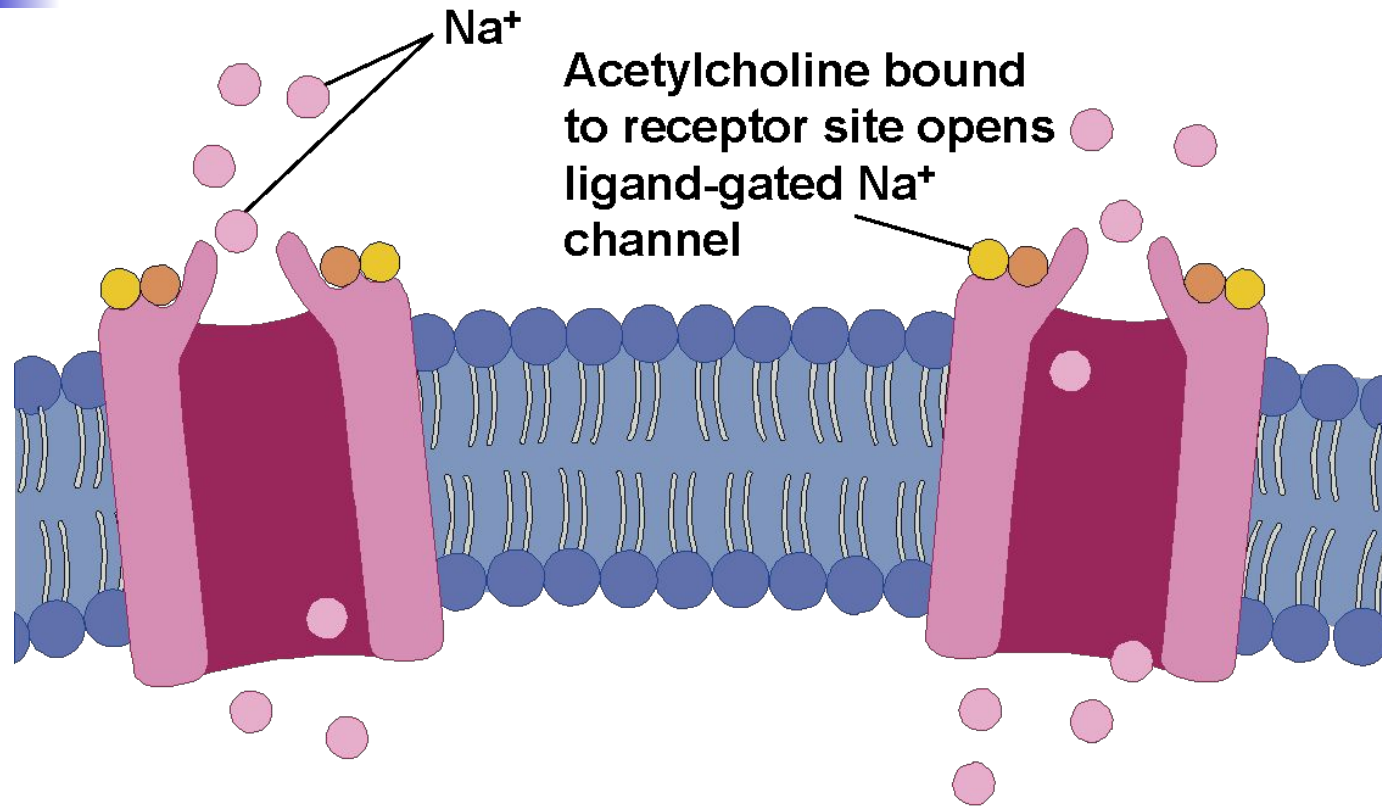
- Motor end plate contains nicotinic receptors for Ach , which are ligand gated ion channels.
- Ach binds to the alpha subunits of nicotinic receptors and causes conformational change.
- When conformational changes occurs ,the central core of channels opens & permeability of motor end plate to Na^+ & K^+ increases.

Sequence Of Events At Neuromuscular Junction

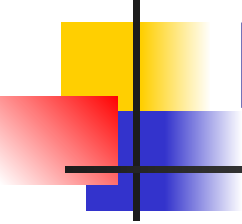
(continued)



Sequence Of Events At Neuromuscular Junction (continued)

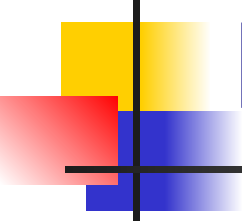


Acetylcholine molecules combine with their receptor sites and cause ligand-gated Na⁺ channels to open.



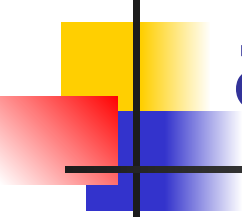
End plate potential

- When the ion channel on post synaptic membrane opens both Na^+ & K^+ flow down their concentration gradient.
- At resting potential net driving force for Na^+ is much greater than K^+ , when Ach triggers opening of these channels more Na^+ moves inwards than K^+ outwards, depolarizing the end plate. this potential change is called end plate potential (EPP).
- EPP is not an action potential but it is simply depolarization of specialized motor end plate.



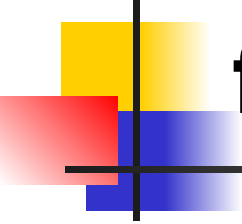
End plate potential (continued)

- Small quanta (packets) of Ach are released randomly from nerve cell at rest, each producing smallest possible change in membrane potential of motor end plate, the MINIATURE EPP.
- When nerve impulse reaches the ending, the number of quanta release increases by several folds and result in large EPP.
- EPP then spread by local current to adjacent muscle fibers which r depolarized to threshold & fire action potential.



Acetyl cholinesterase ends Ach activity at N.M junction

- To ensure purposeful movement ,muscle cell electrical response is turned off by acetylcholinestrerase(AchE), which degrade Ach to choline & acetate.
- About 50%of choline is returned to the presynaptic terminal by Na^+ choline transport to be reused for Ach synthesis.
- Now muscle fiber can relax ,if sustained contraction is needed for the desired movement another motor neuron AP leads to release of more Ach.



Agents & disease that alters the function of N.M junction

- ▢ **Black widow spider venom**: the venom of black widow spider exerts its effect by triggering **explosive release of Ach** from the storage vesicles, not only at N.M junction but all cholinergic sites. All cholinergic sites undergoes prolong depolarization.
- ▢ The most harmful result is respiratory failure.

- **Botulinum toxin:**

botulinum toxin exerts its lethal effect by **blocking the release of Ach** from presynaptic terminals, causing total blockade of neuromuscular transmission, paralysis of skeletal muscle, and, eventually, death from respiratory failure.

- Clostridium botulinum poisoning most frequently result from improperly canned food contaminated with clostridia bacteria.

Physiology:

Nerve action potential (AP)

Ach release

Depolarization (EPP)

Muscle action potential (MAP)

Spread of MAP via TTS

Muscle contraction

0.5-0.8 msec

Myelin sheath

Node of Ranvier

Botulinum toxin

Nicotinic receptor

Mg²⁺

Ach receptors

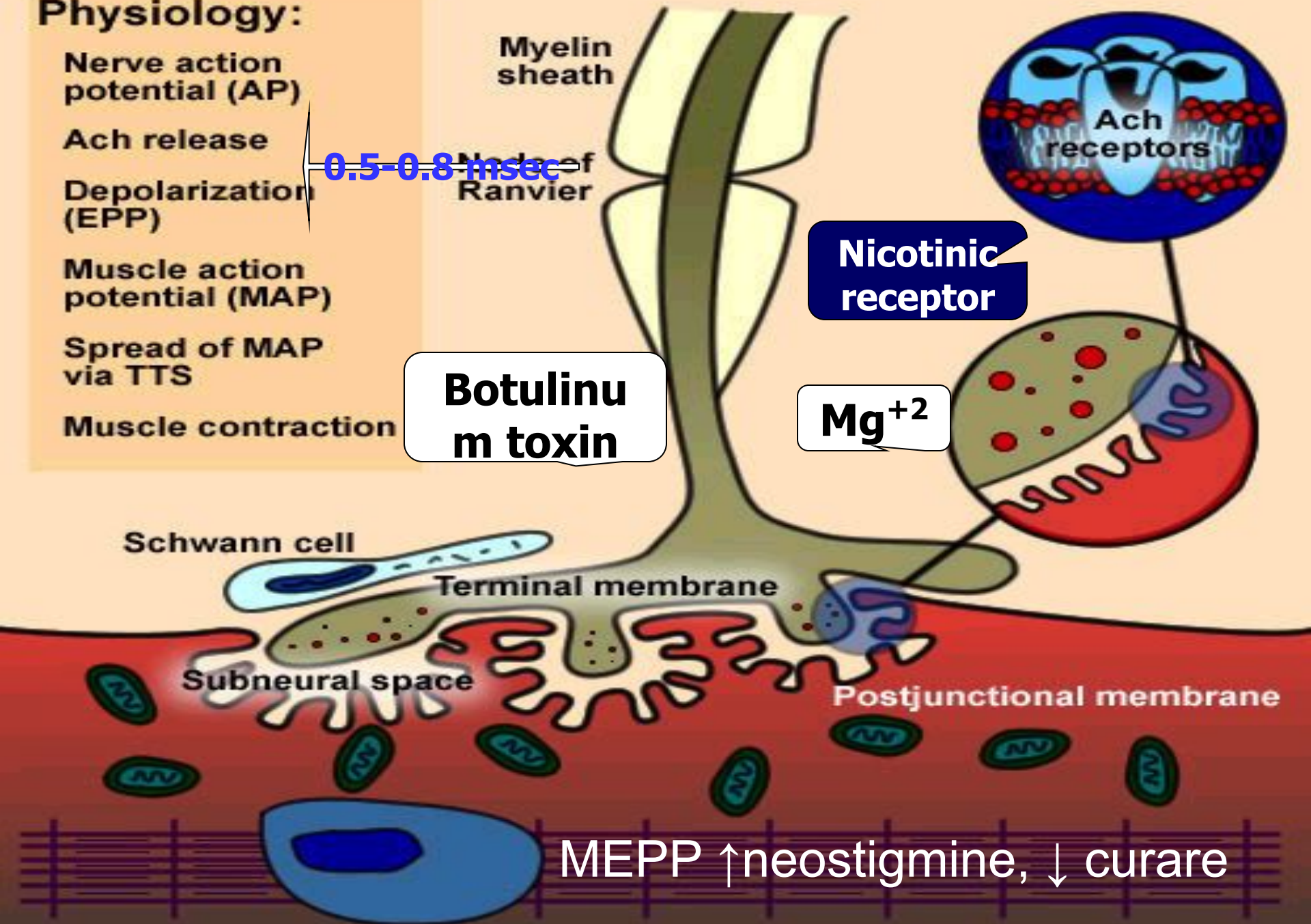
Schwann cell

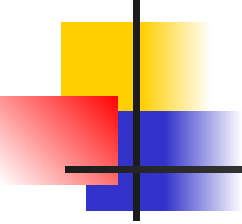
Terminal membrane

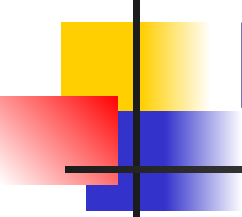
Subneural space

Postjunctional membrane

MEPP ↑ neostigmine, ↓ curare



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- **Curare** :_ curare **competitively binds to Ach receptor** sites on motor end plate ,so Ach can not combine with these sites to open ion channels .and muscles paralysis ensues .
 - In sever poisoning person dies of respiratory failure.
 - In past it was used as deadly arrowhead poison.

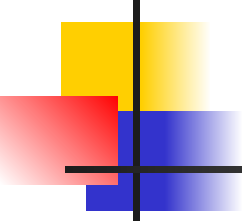


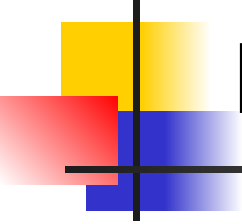
Myasthenia gravis

- A disease involving N.M junction is characterized by the extreme muscular weakness (myasthenia=muscular & gravis=severe)
- It is an auto immune condition (auto immune means immunity against self) in which the body erroneously produces antibodies against its own motor end plate ach receptors.
- Thus not all Ach molecules can find functioning receptors site with which to bind.
- As a results ,AchE destroys much of Ach before it ever has a chance to interact with receptor site & contribute to EPP.

□ Treatment :

it is treated with long acting anticholinesterase inhibitor **pyridostigmine** or **neostigmine**. Which maintains the Ach levels at N.M junction at high levels thus prolonging the time available for Ach to activate its receptors.

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- https://youtu.be/_k6QINRcdV4?si=bgUNxHF9kIKHCnfJ



References

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- Text book of physiology by Linda S. Costanzo, third edition